HAMID KHABIRI, PhD

ROBOTICS ENGINEER | MECHANICAL DESIGN · LOW-LEVEL CONTROL · SYSTEM INTEGRATION

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PROFESSIONAL SUMMARY

Robotics engineer with 8+ years of research and industry experience independently developing custom robotic systems from concept to functional hardware. Hands-on expertise in mechanical design, manufacturing, electromechanical assembly, real-time low-level control, actuator/sensor integration and safety verification. Developed and tested multi-DOF robotic systems, most recently a 3-DOF robot built with SolidWorks, Beckhoff TwinCAT 3 and EtherCAT.

CORE TECHNICAL SKILLS

Mechanical design & manufacturing: SolidWorks; mechanism and transmission design (capstan, differential timing belt); actuator and bearing selection; engineering calculations, materials, tolerances, DFM, CNC manufacturing drawings and 3D-print prototyping.

Robotics control & modeling: Cascaded PID control and tuning; joint- and Cartesian-space control; impedance and admittance control; FK/IK, Jacobians and trajectory planning; dynamic modeling (URDF, Pinocchio) with Coulomb–viscous friction modeling for model-based feedforward.

Real-time automation & software: TwinCAT 3 (Structured Text, PLC/NC, HMI); EtherCAT (PDO/SDO configuration, synchronization and diagnostics); TwinCAT ADS; EPOS Studio; C++/MFC; Unity; MATLAB (data analysis, MSc research).

Electromechanical integration & safety: Maxon BLDC motors and EPOS4 servo drives; absolute encoder and force/torque sensor calibration; wiring, digital/analog I/O, hardware bring-up and fault diagnosis; risk assessment; layered safety: emergency stop, hardware limit switches, STO on the drives, virtual walls and HMI stop.

Engineering testing & verification: Structured robot bench verification, subsystem functional testing, actuator commissioning, control-loop evaluation and hardware troubleshooting.

ENGINEERING EXPERIENCE

Postdoctoral Fellow – Robotics R&D | University of Alberta, Edmonton, Canada Nov 2024–Present

- Independently conceived, designed and built a custom 3-DOF upper-limb robotic system, from requirements and mechanism architecture through fabrication, assembly, electrical integration and testing.
- Designed a capstan-based rotary transmission, a differential timing-belt mechanism and a forearm/wrist end-effector in SolidWorks; completed mechanical calculations, tolerance specifications, material selection, DFM and CNC manufacturing drawings.
- Selected, assembled and integrated three Maxon BLDC motors with EPOS4 EtherCAT drives, transmissions, absolute encoders and force/torque sensing; wired the electrical hardware and calibrated sensor feedback.
- Developed real-time, multi-axis motion control in TwinCAT 3 (Structured Text, PLC/NC) over EtherCAT; configured PDO/SDO mapping and position/velocity/torque modes, and diagnosed fieldbus faults.
- Implemented cascaded control: tuned the EPOS4 inner current, velocity and position loops in EPOS Studio and developed outer-loop joint PID controllers in TwinCAT.
- Implemented FK/IK, Jacobian-based control and trajectory planning; developed impedance control from joint position/velocity feedback and admittance control using force/torque sensor feedback.
- Built a URDF dynamics model of the planar robot in Pinocchio, streamed to TwinCAT via ADS, and combined it with Coulomb–viscous friction modeling for model-based feedforward.
- Developed a TwinCAT HMI and Unity rehabilitation games connected to live robot motion via ADS.
- Performed hazard and risk assessments and implemented layered hardware/software safety: emergency stop, hardware limit switches, Safe Torque Off on the drives, software virtual walls, an HMI stop and fault handling.

- Conducted structured bench verification and subsystem testing; used measured actuator performance and fault investigation to iterate the mechanical and control design, and incorporated preliminary usability observations with five participants and clinician feedback.

Mechanical Designer | Noavaran Sanat Rabin (NSR), Tehran, Iran Apr 2023–Sep 2024

- Designed, fabricated and assembled the mechanical system of a cell-culture incubator with real-time image monitoring, including material and component selection and production planning.
- Developed mechanical components for a stereotactic head frame and performed fabrication planning and assembly.

Mechanical Engineer | Sina Robotics and Medical Innovators, Tehran, Iran Jan 2019–Dec 2020

- Designed and fabricated the mechanical structure and mechanisms of the SinaSim laparoscopic surgery simulator, preparing mechanical calculations, CAD assemblies and manufacturing drawings.
- Assembled, tested, troubleshot and improved mechanical subsystems through iterative prototype development.

SELECTED ROBOTICS PROJECTS

Underactuated Cable-Driven Robotic Hand Orthosis | PhD, Tehran University of Medical Sciences

- Independently designed, built and controlled an active hand orthosis using one motor and underactuated mechanisms for index/middle finger flexion and extension; the wearable section weighed 90 g.
- Developed and experimentally validated a Bowden-cable friction model for fingertip-force estimation using actuator-side sensing (16.5% mean model error over a 0–120° range); used the results in control design.
- Implemented C++/MFC control software for a Maxon motor driver, with an operator interface; carried out testing with a healthy volunteer, two stroke patients and one participant with spinal cord injury.

MRI-Compatible Wrist Robot | Research collaboration, Tehran University of Medical Sciences

- Designed, fabricated and assembled a 2-DOF pneumatic wrist robot for flexion/extension and ulnar/radial deviation using MRI-compatible materials; contributed to pilot fMRI testing and compatibility evaluation.

Wrist–Finger Rehabilitation Robot | Research collaboration, Tehran University of Medical Sciences

- Fabricated and assembled an electromechanical wrist–finger rehabilitation system that was tested with post-stroke participants.

EDUCATION

PhD, Medical Robotics | Tehran University of Medical Sciences, Tehran, Iran — 2022

Thesis: Design, prototyping and clinical evaluation of an active assistive robotic orthosis for hemiplegic patients' hand.

MSc, Mechanical Engineering (Biomechanics) | Sahand University of Technology, Tabriz, Iran — 2014

Thesis: The effect of eccentric exercise on isometric contraction of knee extensor muscles.

BSc, Biomedical Engineering (Biomechanics) | Islamic Azad University, Science and Research Branch, Tehran, Iran — 2011

PATENTS & INTELLECTUAL PROPERTY

- U.S. Patent Application US20260109022A1, "A Wearable Assistive Robot" (published 2026) — named inventor.
- Iranian Patent No. 101489 — Active assistive hand orthosis (2020) — named inventor.
- Iranian Patent No. 104035 — Hand, wrist and finger physiotherapy robot (2021) — named inventor.
- Iranian Patent No. 96394 — MRI-compatible wrist rehabilitation robot (2018) — named inventor.
- University of Alberta Invention Disclosure D2027-0005 — Upper-limb rehabilitation robotic system — named inventor.

PEER-REVIEWED PUBLICATIONS

1. F. Mohandesi, A. Mirbagheri, **H. Khabiri**, M.M. Mirbagheri, N.N. Ansari, M. Norouzi, R. Kazemi, M. Aghanouri. "Design and Development of a Robotic System for Hand's Wrist-Fingers Rehabilitation." Scientia Iranica, 32(2), 2025. doi:10.24200/sci.2023.59650.6362.

2. M.R. Hosaini, **H. Khabiri**, H. Sharini, V. Dehlaghi, A. Safarpoor. “Development and Validation of Virtual Reality Combined with Shoulder Wheel Device for Active Rehabilitation Training.” International Journal of Biomedical Engineering and Technology, 46(2), 116–137, 2024. doi:10.1504/IJBET.2024.141570.
3. H. Sharini, N. Riyahi Alam, **H. Khabiri**, H. Arabalibeik, H. Hashemi, A.R. Azimi, S. Masjoodi Javadi. “Novel fMRI-Compatible Wrist Robotic Device for Brain Activation Assessment During Rehabilitation Exercise.” Medical Engineering & Physics, 83, 112–122, 2020. doi:10.1016/j.medengphy.2020.05.008.
4. H. Sharini, S. Zolghadriha, N. Riyahi Alam, M. Jalalvandi, **H. Khabiri**, H. Arabalibeik, M. Nadimi. “Assessment of Motor Cortex in Active, Passive and Imagery Wrist Movement Using Functional MRI.” Journal of Biomedical Physics & Engineering, 11(4), 515–526, 2021. doi:10.31661/jbpe.v0i0.1034.
5. Z. ZangiAbadi, **H. Khabiri**, A. Mirbagheri, G. Halvani, M. Askarishahi, M. Nasiri. “Investigation of Musculoskeletal Symptoms, Work Postures, Quantification of Muscle Activity, and Estimation of Grip/Push Forces Among Sonographers.” Human Factors and Ergonomics in Manufacturing & Service Industries, 34(6), 589–600, 2024. doi:10.1002/hfm.21051.
6. M. Nasiri, **H. Khabiri**, G.H. Halvani, M. MoradiAmin, F. Shamsi, Z. ZangiAbadi. “Anthropometric Analysis of the External Ear Using CT Scan Image by Age, Gender, and Side Among the Iranian Population.” Ergonomics in Design, 33(2), 89–106, 2025. doi:10.1177/10648046241228360.
7. M. Azghani, S. Kian-Bostanabad, T. Ahmadi, **H. Khabiri**. “Evaluation of Bone Butterfly Fracture by Controlling the Applied Load at Different Strain Rates.” Iranian Journal of Biomedical Engineering, 2016.
8. **H. Khabiri**, M.N. Talebi, M.F. Kamran, S. Akbari, F. Zarrin, F. Mohandesi. “Music-Induced Emotion Recognition Based on Feature Reduction Using PCA From EEG Signals.” Frontiers in Biomedical Technologies, 11(1), 59–68, 2024. doi:10.18502/fbt.v11i1.14512.

MANUSCRIPTS & PRESENTATIONS

Under review: H. Khabiri, M. Nazarahari. “A Comprehensive Review of End-Effector Rehabilitation Robots with a Focus on the Shoulder and Elbow Joints.” Robotics and Autonomous Systems (2026).

Conference poster: H. Khabiri, M. Nazarahari. “A Review of End-Effector Rehabilitation Robots for Shoulder and Elbow Joints.” RISEx 2025, Edmonton, Canada.

TEACHING & MENTORSHIP

- Adjunct Lecturer, Shahid Sadoughi University of Medical Sciences (during PhD): Occupational Biomechanics (5 semesters), Ergonomics in Design (4), Anthropometry (1) and Solid Mechanics (1).
- Teaching Assistant: Robot Kinematics and Dynamics, Tehran University of Medical Sciences (2016); Statics, Sahand University of Technology (2013).
- Mentored undergraduate and visiting students at the University of Alberta (2025–2026); co-supervised MSc research in ergonomics and anthropometry resulting in peer-reviewed publications.

AWARDS & RECOGNITION

- Selected participant, Lab2Market Health: Validate (2026) — customer discovery and commercialization training.
- Selected Researcher, Iran’s National Elites Foundation (2023).
- Ranked 2nd nationally in the PhD entrance examination (2015).
- Ranked 3rd in MSc cohort, Sahand University of Technology (2014).